Sebastian Fernandez Del Castillo

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EDUCATION

Bachelor of Science in Mechanical Engineering — The University of Texas at Austin
Expected Graduation: December 2027 — GPA: 3.4

SKILLS

- **Software:** SolidWorks, Fusion 360, CAM, Finite Element Analysis (FEA), MATLAB, Arduino
- **Manufacturing:** 3D Printing, Laser Cutting, CNC, Manual Mill, Manual Lathe, Welding, Soldering, GD&T
- **Languages:** English, Spanish (Fluent)
- **Work Eligibility:** U.S. Citizen

TECHNICAL EXPERIENCE

ASCEND Texas - GoAERO & XPRIZE Wildfire Challenge

Production Lead — *May 2025 - May 2026*

- Led production integration for ASCEND's hybrid aerial-terrestrial robot across avionics, actuation, steering, and propulsion
- Guided linear actuator mounting from standalone parafoil-steering setup to ATV integration for steering tests
- Coordinated actuator mount **CAD** reviews and **3D-printed** prototypes around frame clearance, travel, and support constraints
- Developed **Fusion 360 CAM** workflow and **CNC** operations for steering hardware, enabling physical subsystem testing
- Trained 5-member production team in shop safety, **manual mill/lathe** basics, and **welding** practice for fabrication readiness

Mechanical Design & Production Engineer — *Jan 2025 - May 2025*

- Designed and fabricated **laser-cut** avionics enclosure in **SolidWorks** for flight electronics and batteries
- Created component layout plan around EMI, airflow, wiring access, and component-specific placement constraints
- Refined panel fit, ventilation, cable routing, and removable access to improve manufacturability and assembly
- Coordinated with mechatronics to integrate servo-switch hardware and dedicated batteries for avionics and actuation

RC Racecar - *Machine Elements Design* — *Aug 2025 - Dec 2025*

- Led fabrication and assembly of RWD RC racecar with **laser-cut** chassis, **3D-printed** steering, and ~10:1 gearbox
- Applied team gear-ratio, torque-speed, and chassis **FEA** analysis to guide drivetrain selection and durability checks
- Completed three-lap race in 21.05 seconds, identifying tire traction and shaft-collar reliability as key limits

Texas Spacecraft Laboratory - *REACT Hardware Engineer* — *Jan 2025 - Present*

- Designed **SolidWorks** layouts for reaction-wheel magnetic-field testing with magnetometers, cameras, and electronics
- Evaluated non-ferromagnetic material and sensor-placement constraints to reduce interference in magnetometer testing
- Developed optical-table and slider-based camera mounting concepts to improve test visibility
- Evaluating motor and Helmholtz coil cooling concepts around thermal and magnetic-interference constraints

Cam-Driven Drawing Machine - *Robotic Mechanism Design* — *Aug 2025 - Dec 2025*

- Built and tested cam-driven drawing machine converting one motor input into coordinated X/Y and pen-lift motion
- Assembled cam, follower, slider, spring, and linkage systems to translate rotary motion into controlled pen movement
- Worked with team on **position/velocity analysis** to guide cam profiles, slider travel, and pen-lift timing

Advanced Robotic Technologies for Surgery (ARTS) Laboratory - *Student Researcher* — *May 2025 - Jul 2025*

- Analyzed flexible surgical screw bending data from **LUNA fiber-optic sensors** and **VICON motion capture**
- Tuned **MATLAB** inputs for origin, tip position, and marker intervals to align LUNA reconstructions with VICON data
- Compared bending accuracy using MATLAB plots of strain, curvature, and tip/shape error across free, sharp, and round cases

LEADERSHIP EXPERIENCE

Cubes4Good - *Founder* Website: https://cubes4good.org Instagram (@cubes4good) — *May 2022 - Present*

- Founded Cubes4Good to teach Rubik's cube techniques to children undergoing long-term medical treatment or disabilities, supporting cognitive engagement and emotional well-being
- Developed instructions, built a website, and raised donations to provide free cubes during workshops, reaching 150+ children and hospital TV broadcasts in 600+ rooms
- Led workshops in Houston, Dallas, and Mexico City with MD Anderson, Texas Children's Hospital, Children's Health, Ronald McDonald House Charities, and Casa de la Amistad